

TUTORIAL – 7

Q1. In SHA-256, Given $x=1$, what is $\Sigma_o(x), \sigma_0(x)$?

$\oplus$	bitwise XOR
$\wedge$	bitwise AND
$\vee$	bitwise OR
$\neg$	bitwise complement
$+$	mod 2^{32} addition
R^n	right shift by n bits
S^n	right rotation by n bits

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z)$$

$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z)$$

$$\Sigma_0(x) = S^2(x) \oplus S^{13}(x) \oplus S^{22}(x)$$

$$\Sigma_1(x) = S^6(x) \oplus S^{11}(x) \oplus S^{25}(x)$$

$$\sigma_0(x) = S^7(x) \oplus S^{18}(x) \oplus R^3(x)$$

$$\sigma_1(x) = S^{17}(x) \oplus S^{19}(x) \oplus R^{10}(x)$$

Q2. Consider a mini version of ECDSA based on the elliptic curve $E: y^2 = x^3 + x + 6$ over $GF(p)$ where $p=11$, the group generator is $A=(2,7)$ and the order of the group is 13.

- If the private key $x=7$, what is the public key?
- Given the hash value $h(M)=10$ of a message M and choose $k=3$, compute the ECDSA signature on the message M and verify the signature.

$$\sum_1(x) = S^6(x) \oplus S^{11}(x) \oplus S^{25}(x)$$

$$\sum_0(x) = S^2(x) \oplus S^{13}(x) \oplus S^{22}(x)$$

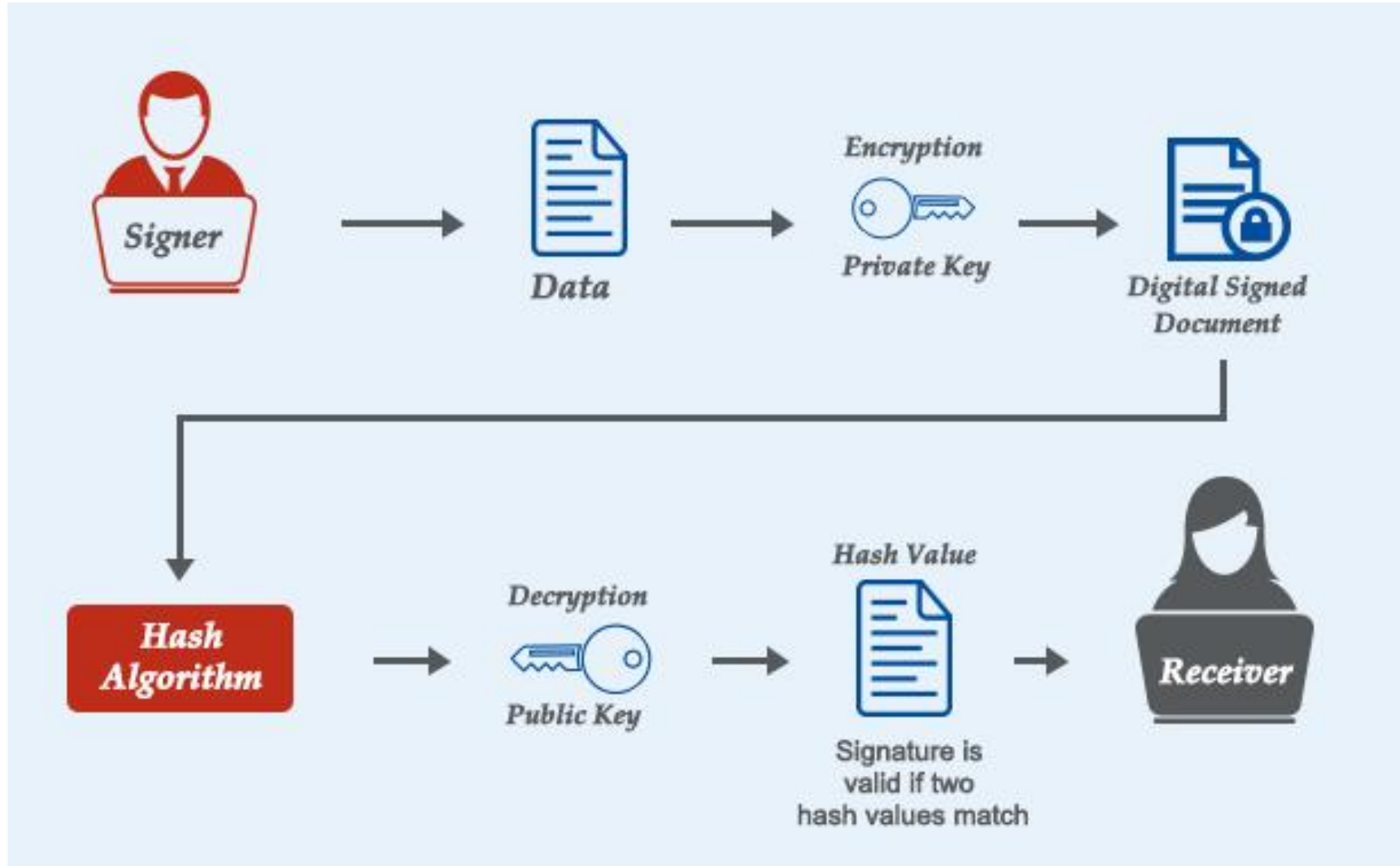
[illegible]

$$\sigma_0(x) = S^7(x) \oplus S^{18}(x) \oplus R^3(x)$$

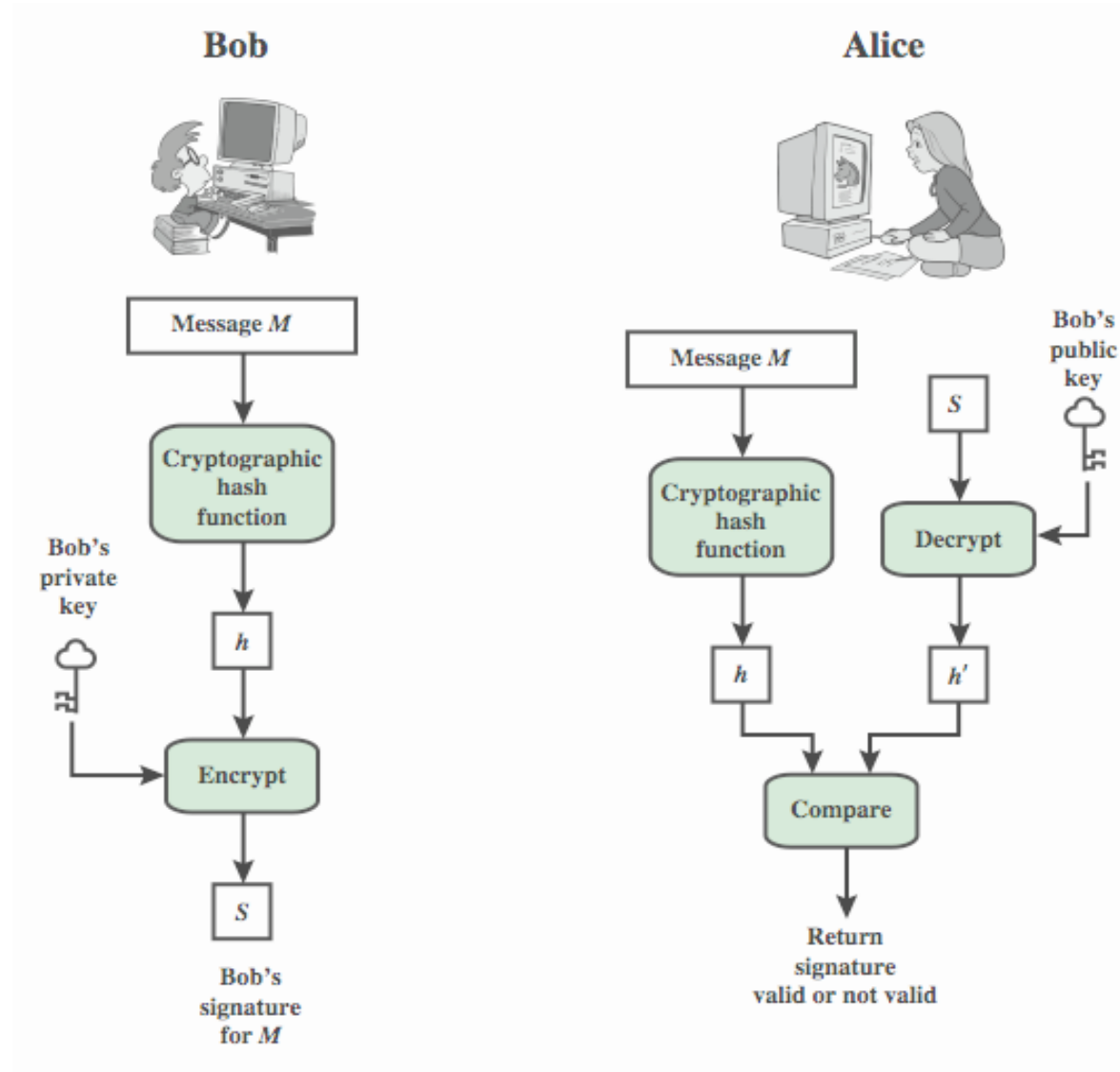
$$\sigma_1(x) = S^{17}(x) \oplus S^{19}(x) \oplus R^{10}(x)$$

[illegible]

Digital Signature



Digital Signature Verification



Elliptic Curve DSA (ECDSA)

- Let p be a prime, and let E be an elliptic curve defined over $GF(p)$. Let α be a point on E having prime order q , such that Discrete Logarithm problem in $\langle \alpha \rangle$ is infeasible.

Define $K = \{ (p, q, E, \alpha, x, Y) : Y = x\alpha \}$

p, q, E, α, Y are the public key, x is private

ECDSA Contd.

- **Signing:** For a (secret) random number k , define $sig_x(m, k) = (r, s)$, where $k\alpha = (u, v)$, $r = u \bmod q$ and $s = k^{-1} (Hash(m) + xr) \bmod q$
- **Verification:** For a message $(m, (r, s))$, verification is done by performing the following computations:
$$i = Hash(m) \bullet s^{-1} \bmod q$$
$$j = r \bullet s^{-1} \bmod q$$
$$(u, v) = i\alpha + jY$$
$$ver(m, (r, s)) = true \text{ if and only if } u \bmod q = r$$

ECDSA Example

E: $y^2 = x^3 + x + 6$ over GF(11), the Generator is $A=(2,7)$ and the Order is 13.

If $X = 7$ what is $Y = ?$

If $h(M)=10$ of M and $k=3$ then Verify the Sign

Signing:

$$kA=3(2,7)=(8,3) = (u,v), u=8, v = 3$$

$$r=u \pmod{q}=8 \pmod{13}= ? \quad [\text{sig}_x(m, k) = (r, s)] \quad [r = u \pmod{q}]$$

$$s = k^{-1} (\text{Hash}(m) + xr) \pmod{q}$$

$$s=k^{-1}(h(M)+xr) \pmod{q} = 3^{-1}(10+7 \times 8) \pmod{13} = ?$$

So, the ECDSA signature on M is calculated (r, s)

ECDSA Example Contd.

E: $y^2 = x^3 + x + 6$ over GF(11), the Generator is $A=(2,7)$ and the Order is 13.

If $X = 7$ what is $Y = ?$

If $h(M)=10$ of M and $k=3$ then Verify the Sign

Verifying:

Public Key, $Y = xA = 7(2,7) = (7,2)$

$$[(u, v) = i\alpha + jY]$$

$$i = h(M) s^{-1} \pmod{q} = 10 \times 9^{-1} = 10 \times 3 = 4 \pmod{13}$$

$$j = rs^{-1} \pmod{q} = 8 \times 9^{-1} = 8 \times 3 = 24 = 11 \pmod{13}$$

$$iA + jB = 4(2,7) + 11(7,2) = (10,2) + (2,4) = (8,3) = (u, v)$$

$$[r = u \pmod{q}]$$

IF found r is same as calculated r then it is valid